Vishal Rai

Software Engineer | Java, Spring Boot, Microservices, AWS, Kubernetes

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PROFESSIONAL SUMMARY

Software engineer with 3+ years of experience building the platform under FinTech applications. Designs and builds a Spring Boot microservice framework that business teams build on, with authorization, multi-tenancy, batch processing, notifications, configuration and an API gateway built in. Runs it on AWS and Amazon EKS with Helm, Docker and infrastructure as code, ships it through Jenkins and GitLab pipelines, and observes it with Grafana or ELK. Focused on removing manual effort through database release automation, release tooling, MCP servers and AI-skill driven migrations, with measured results of 30-65% less time or effort.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, Bash, Groovy
Backend: Spring Boot, Spring Security, Spring Cloud (Config, Gateway, Consul), Spring Batch, Spring Data JPA, OAuth2 / OIDC, JWT, Resilience4j, Apache Camel, JMS
Data: PostgreSQL, Redis, Elasticsearch, Liquibase
Cloud & DevOps: AWS (CloudFormation, EKS, ECR, S3, EC2, RDS, EFS), Kubernetes, Helm, Docker, Jenkins, GitLab CI, Nexus, HashiCorp Vault
Observability: Grafana, Prometheus, Loki, Tempo, Alloy, ELK Stack (Elasticsearch, Logstash, Kibana), OpenTelemetry
Frontend: React, Angular, Module Federation, Redux Toolkit, SlickGrid
Testing & Tools: JUnit, Mockito, EvoSuite, Randoop, Diffblue Cover, Git, Jira, Confluence, JasperReports, FineReport
AI & Automation: MCP servers, Claude Code, AI agent skills, XyOps, JS7, job2do

PROFESSIONAL EXPERIENCE

Associate Software Developer, NRIFT India Pvt Ltd Aug 2023 - Present
Kolkata, India | Grade 1 Level 2 since Apr 2025; Grade 1 Level 1, Aug 2023 - Mar 2025.

Backend & Microservice Frameworks

- Implemented menu- and action-level authentication and authorization on an in-house authorization server and Auth0, using Spring Security, JWT, role-derived authorities and an ordered security filter chain.
- Designed multi-tenancy into the framework, with tenant context, automatic data segregation and tenant-scoped access control inherited by every adopting application.
- Built a reusable Spring Batch framework where a batch is wired with one descriptor, with automatic restart and rerun handling and distributed execution through remote partitioning over messaging.
- Built an API gateway with custom filters and Redis-backed rate limiting by IP, user and API, Consul service discovery and Resilience4j circuit breaking.
- Implemented profile-based configuration with a central config server and HashiCorp Vault for secrets, using least-privilege policies.
- Designed an event-driven email notification framework over SMTP with a shared payload contract and delivery log.

Cloud, Kubernetes & DevOps

- Deployed the microservice platform on Amazon EKS from cluster creation onward, with Helm charts for every workload and manifests maintained across Minikube (DEV, SIT) and EKS (UAT, PROD).
- Implemented domain and child-domain routing with the NGINX ingress controller, cert-manager and AWS load balancers, HA replicas, log rotation and namespace segregation; a resource report across about 30 pods supported right-sizing and cost savings.
- Codified a multi-client AWS estate as parameterized CloudFormation templates, cutting new environment creation time by over 40%.
- Built a Python platform on botocore and the CloudFormation IaC Generator that reverse-engineers live environments into templates, cutting lookalike environment creation time by 60%.

- Built CI/CD pipelines from scratch with Jenkins (Groovy) and GitLab Runner, and centralized Docker deployments from Nexus, S3, ECR and GitLab registries with versioned, traceable artifacts.
- Integrated switchable Grafana (Prometheus, Loki, Tempo, Alloy) and ELK observability stacks over standardized telemetry.

Automation, Release & AI Engineering

- Led the proof of concept and production adoption of Liquibase across a monolith and microservices, with Maven archetypes for new services, gated migrations and two-tier (feature and release) rollback.
- Built Python release tooling for release scoping across code bases, approved-versus-committed change reconciliation and fix-version promotion, plus daily job anomaly reports.
- Upgraded the framework to Spring Boot 4 using AI agent skills, cutting upgrade effort by 60%, and migrated its workflow engine from Camunda to Operaton.
- Migrated financial job charts from job2do and JS7 to XyOps using AI skills and agent-driven runs, cutting migration and testing effort by over 65%, and built a highly available XyOps scheduler.
- Built two MCP servers adopted organization-wide, behind a shared gateway with directory-based authentication, and a five-stage agentic development lifecycle with human checkpoints.

Frontend, Reporting & Quality

- Contributed to migrating an Angular monolithic frontend to a React micro-frontend framework on Module Federation, and set up Nexus, ECR and GitLab package registries for frontend artifacts.
- Built a widget and workspace data-visualization framework on SlickGrid and chart libraries, kept current by Logstash-based change data capture.
- Centralized JasperReports and FineReport servers with distributed, highly available execution, processing bulk reports 30% faster.
- Automated JUnit generation for a legacy monolith with Mockito after evaluating Randoop, EvoSuite, Amazon Q Developer and Diffblue Cover.

Associate Software Engineer Intern, NRIFT India Pvt Ltd Oct 2022 - May 2023
Kolkata, India

KEY RESULTS

- Over 65% less migration and testing effort moving financial job charts to XyOps.
- 60% less effort for a major Spring Boot upgrade, packaged as reusable skills for other teams.
- Over 40% faster new AWS environment creation, and 60% faster lookalike environments.
- 30% faster bulk report processing through distributed, highly available execution.

EDUCATION

Bachelor of Technology in Computer Science Engineering, Techno Main Salt Lake 2019 - 2023
WBBHSE (Class 12th equivalent), Patha Bhavan 2017 - 2019

LANGUAGES

English (Professional), Hindi (Professional), Bengali (Fluent)